

The artifact originally was the final project of a Data Structures and Algorithms course in which we were deciding which data structure would be the most efficient for taking a csv file and sorting it into a list of courses offered at a school. There are functions with a Binary Search Tree for deletion, insertion, and addition to the tree. There is also a search function for finding a specific course. Finally, there's a function for printing all available courses.

The artifact is included because it shows an understanding of four different data structures and the most common algorithms needed for using the data structures. It was improved by creating an easy to follow and understand graphic that explains that four different types of data structures considered for this project, what they are used for and what they mean, an explanation of the difference in timing for the algorithms, and the attached project so the Binary Search Tree is shown in action. I also included the coding for a Vector, Hash Table, and Linked List that could be modified to replace the Binary Search Tree in the main project file.

This enhancement shows the course outcomes of designing, developing, and delivering professional-quality written and visual communications along with helping to build a collaborative environment as the graphic is meant for those who might be less familiar with computer science but want to understand why things are being programmed a specific way. This enhancement also touches on the idea of security in relation to what data structures are more secure for storing information.

I am considering adding an element to the graphic with oral communications on the graphic explaining it more and going through the attached project to enhance it further.